

2.007 Oral Design Review

Merlin Gogolin

EVERYTHING MUST BE MADE AS SIMPLE AS POSSIBLE, BUT NOT SIMPLER

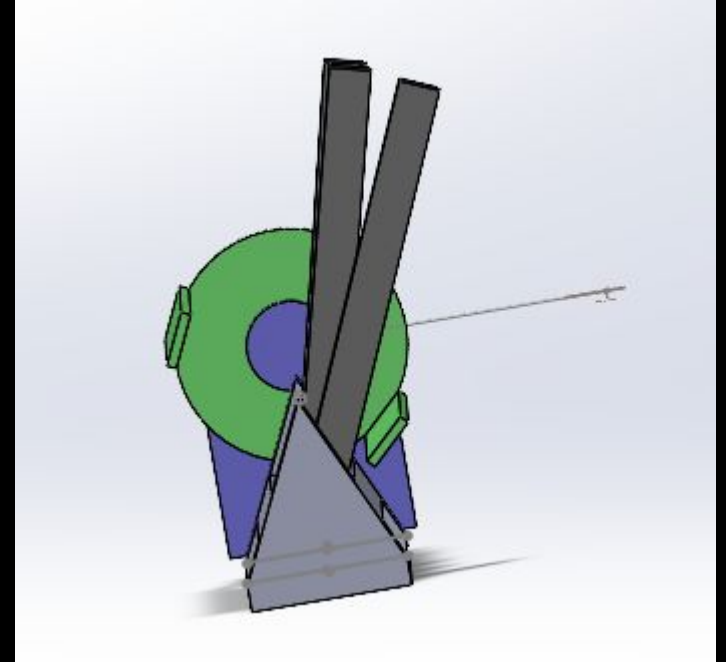
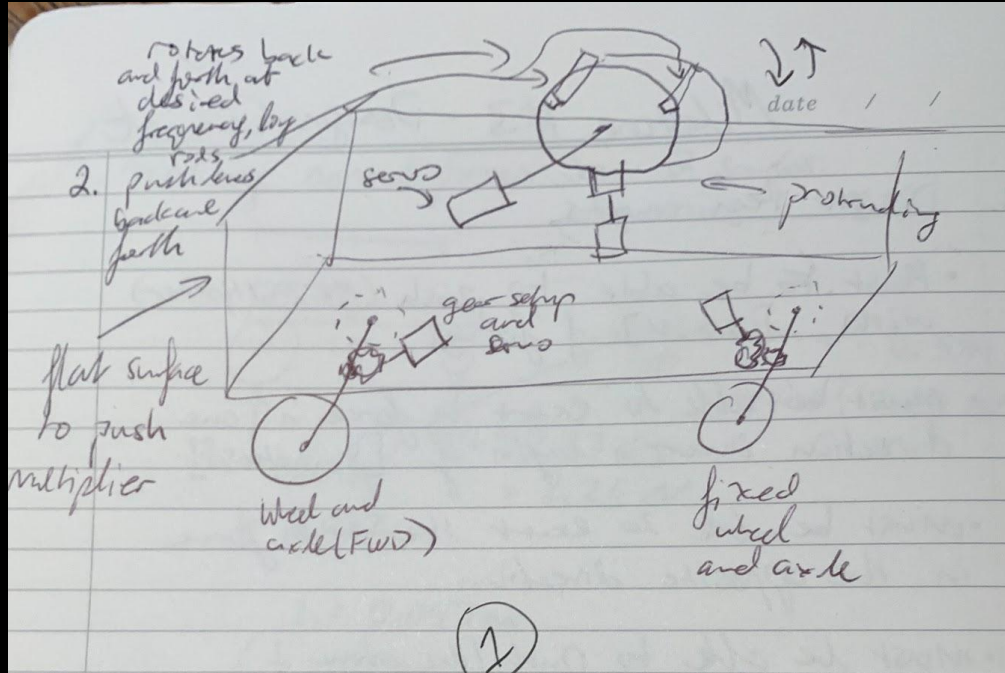
Albert
Einstein

Stage

The Levers

1

The Plan

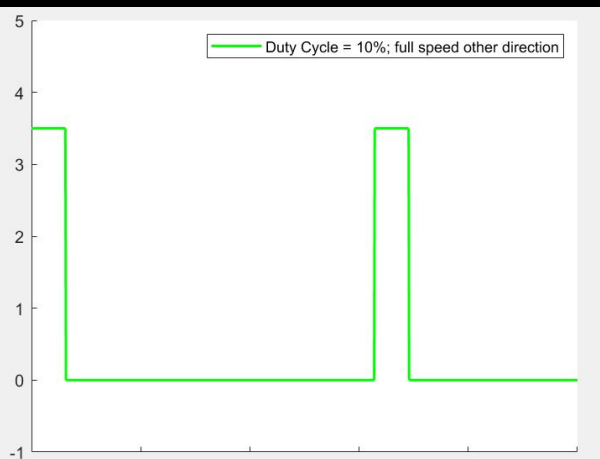
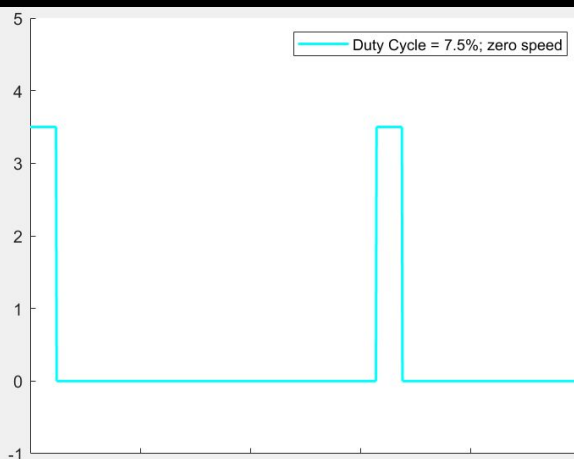
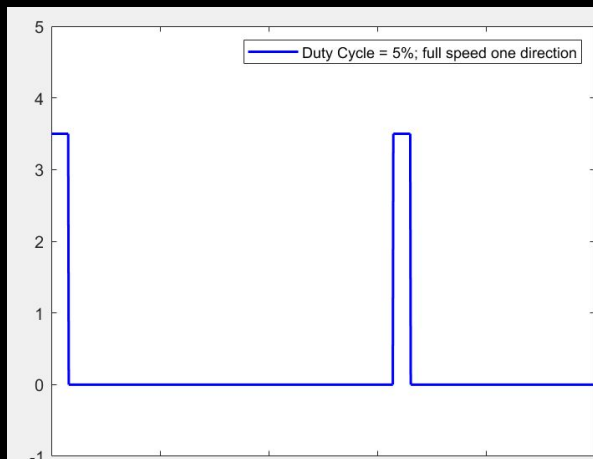


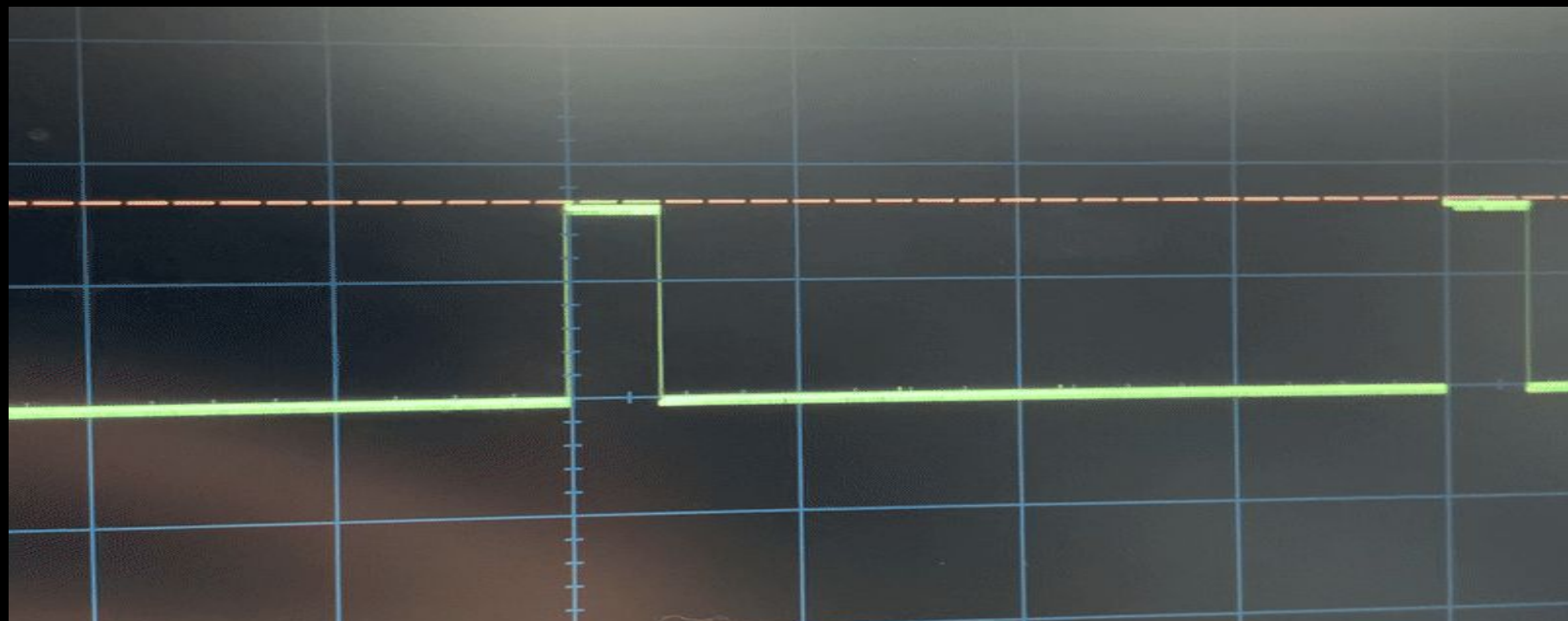
The Problem

```
serv.write(90);  
delay(1000);  
serv.write(0);  
delay(1000);
```

How can you trigger an **oscillation** from the remote control?

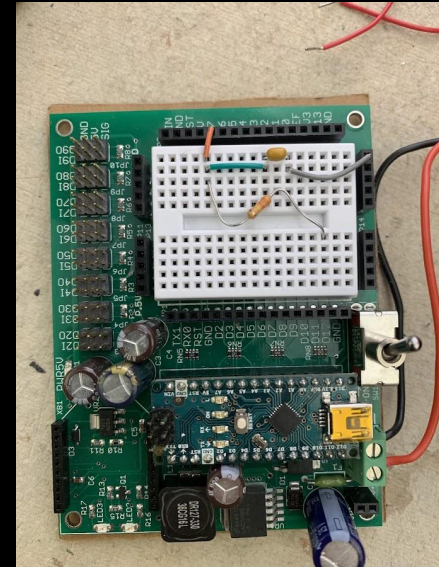
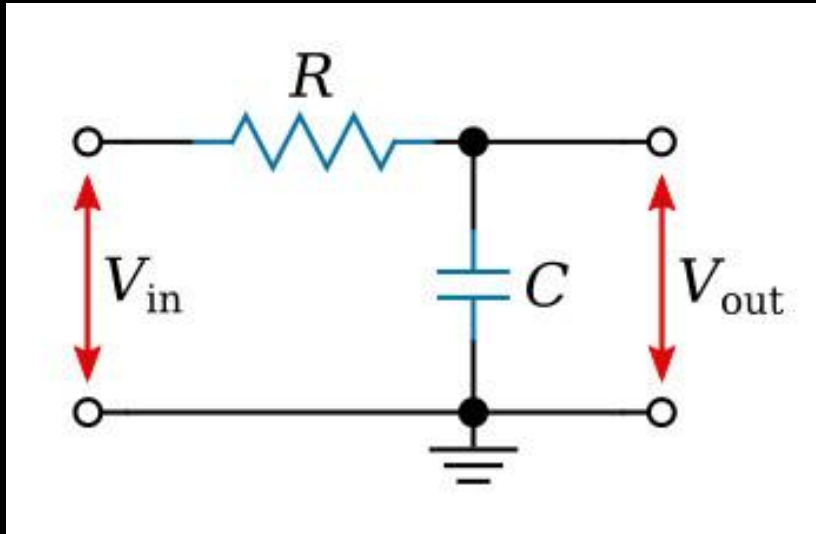
- The oscillation itself is not so difficult to code in Arduino—but how to activate oscillating mode?
- The remote control outputs a **pulse wave** to the servos. This means the signal is either **on** (3.5V) or **off** (0V). What varies is the **percentage of time the signal is on**, or the **duty cycle**.



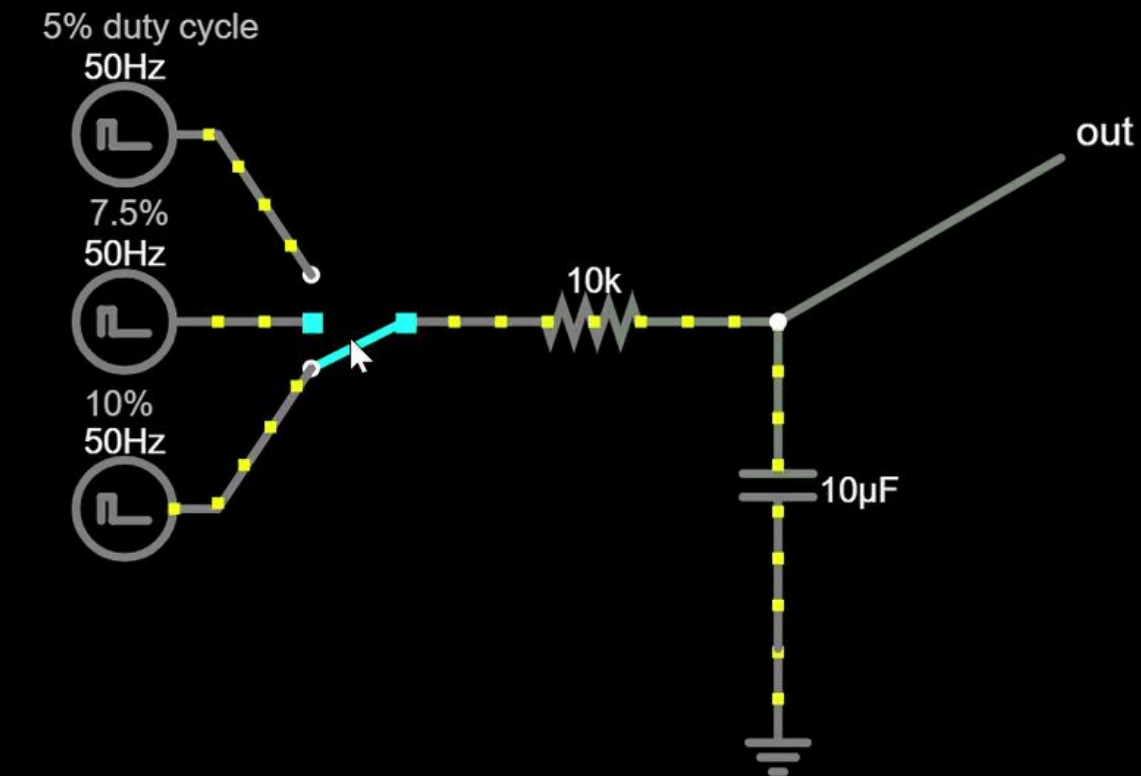


The Solution

Implement an **RC Low-Pass Filter** to convert the **pulse wave** into something very close to a **steady-state analog voltage**, which can then be easily read by Arduino via `analogRead()`*.

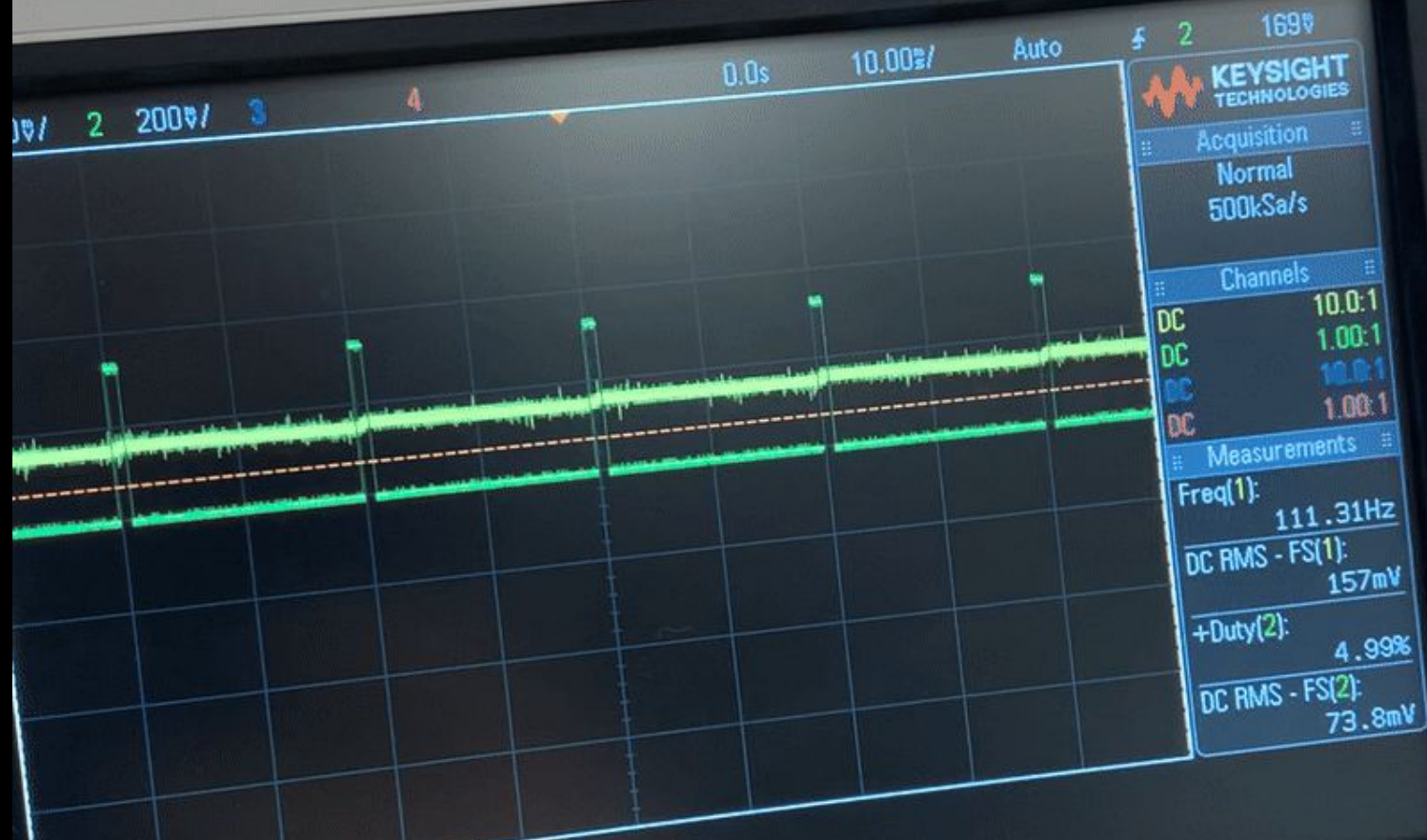


* `analogRead()` reads an input voltage between 0 and 5V on a scale of 0-1023



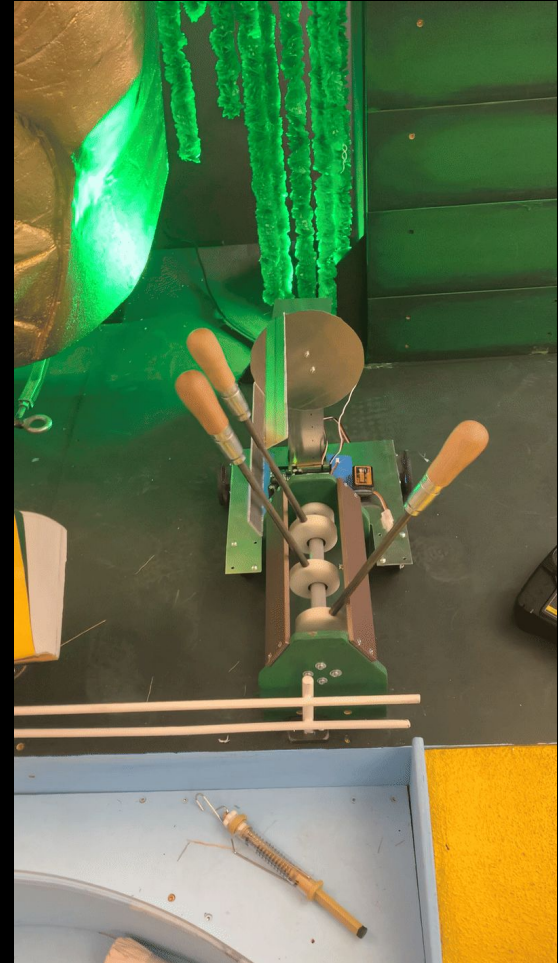
* not to scale





The Problem (2.0)

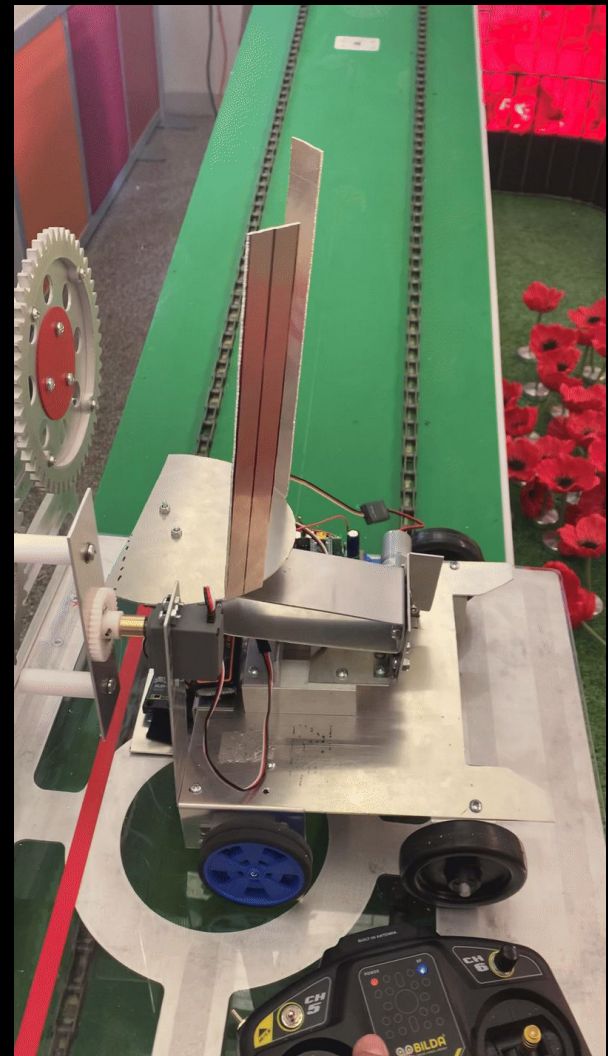
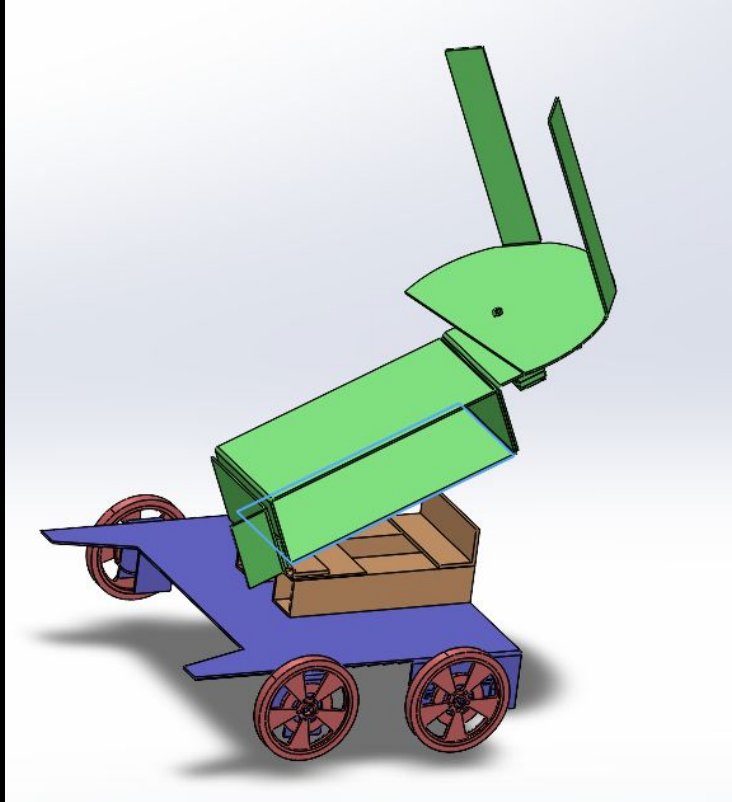
This is when I realized I had failed to consider the size constraint—with the arms, my robot was much too large for the box. The initial, easy fix was to make the whole chassis a U-shape, but this didn't allow for a lot of stability or further additions of modules. Further innovation was needed...creativity was necessary...



"Life imitates art..." - Oscar Wilde



The Solution (2.0)



Stage

The Train

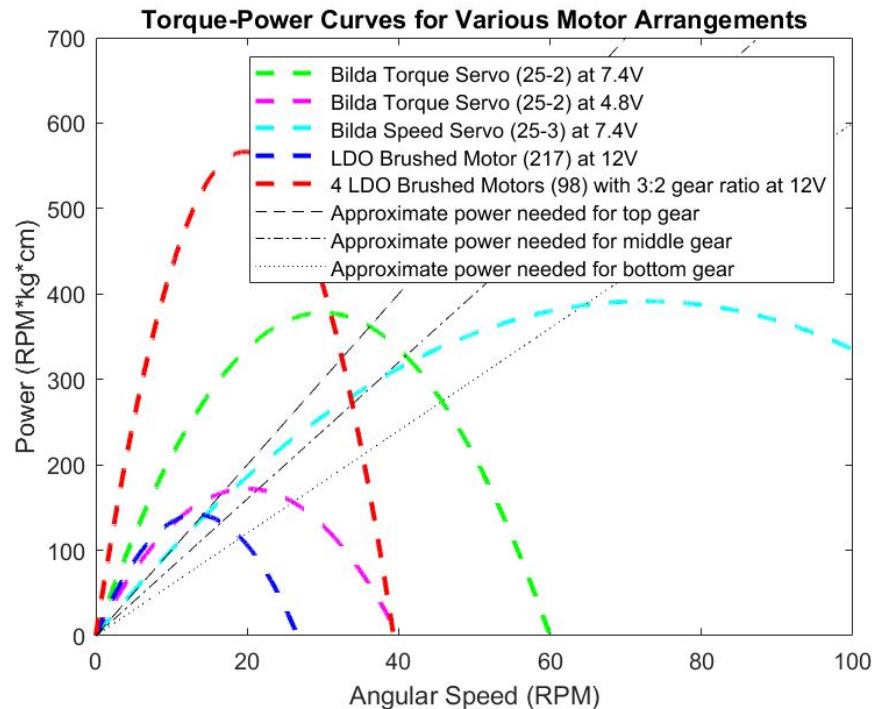
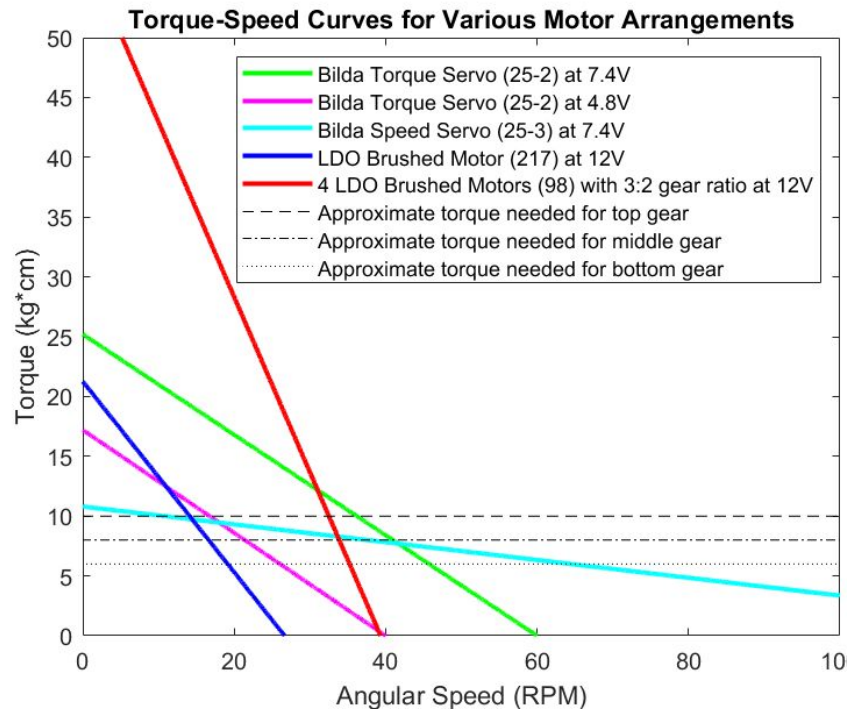
2

The Question(s)

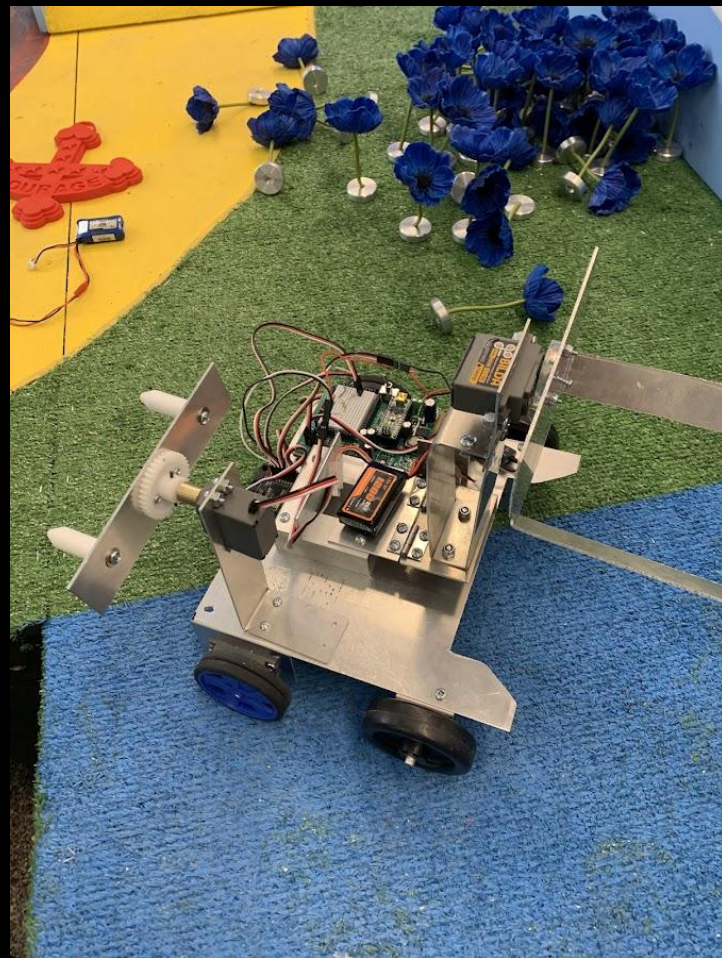
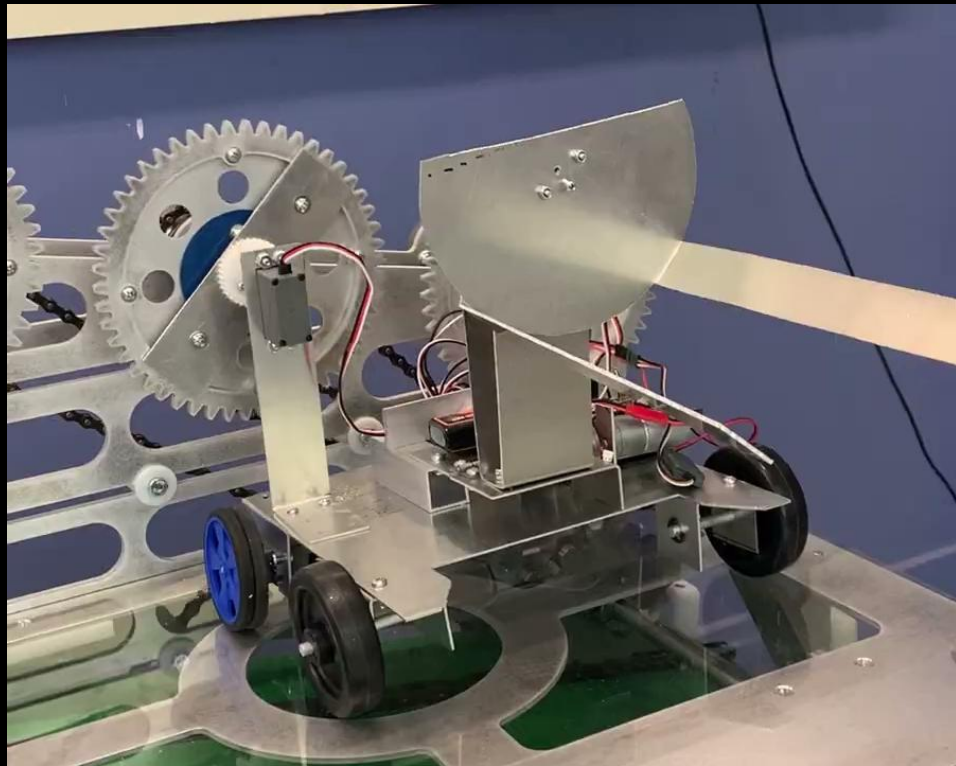
The levers are good, but they don't double for the autonomous period, and driving/parking would be quite difficult to code autonomously, consistently.

- ❖ Can I fit a small, compact, simple-to-build, yet effective spinning system to drive the train within the autonomous period?
- ❖ How does the strength/power of the LDO Brushed Motors compare with that of the Bilda Servos?

The Math



The Execution



Stage

The Multiplier

3

The Future

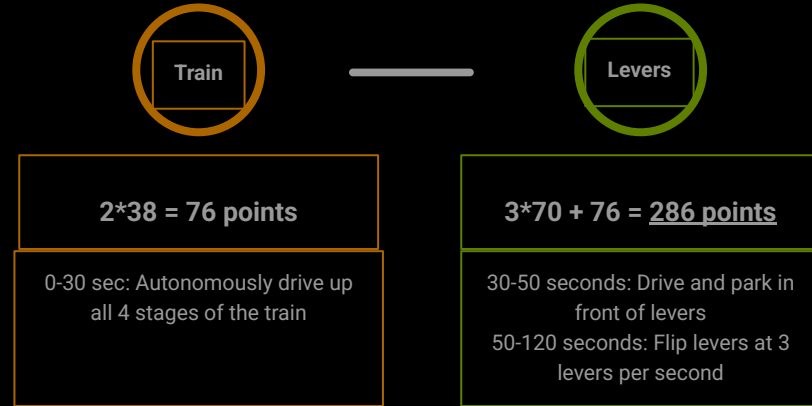
To maximize points, I definitely want to utilize the multiplier. However my resources—time of play, space on my robot, approaching deadlines—are limited. Some options:

- ❖ Drive around to other side and push it
- ❖ Latch on to it and pull it
- ❖ Latch on to it and use Servo to pull it*

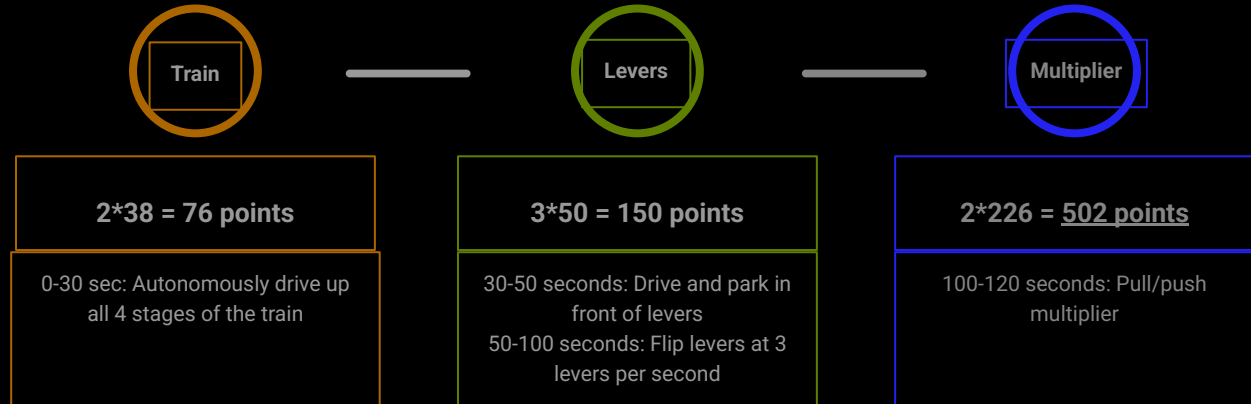
Another task I need to complete is coding the autonomous portion, of driving up the train. For this, I likely will need separate Arduino scripts for both sides.

The Breakdown

Without multiplier:



With multiplier:



The Lessons

- ❖ Do the **math** before doing the building
- ❖ Think **outside the box**
- ❖ Prototype **everything**
- ❖ Build **modular** designs
- ❖ Sometimes the **simplest** solutions are the best solutions
- ❖ Learned how RC and Servos operate!

WE CANNOT SOLVE OUR PROBLEMS WITH THE SAME THINKING WE USED WHEN WE CREATED THEM

Albert
Einstein